

ANN COURSE PROJECT

Main Concepts of **Probability**

Used in the Course

What We'll Cover

01

Foundations

Sample space · Events · Axioms of Probability

02

Probability Rules

Complement · Addition · Multiplication rules

03

Conditional Probability & Bayes

$P(A|B)$ · Total Probability · Bayes' Theorem

04

Random Variables & Distributions

PMF · PDF · CDF · Bernoulli · Binomial · Gaussian

05

Expectation & Variance

$E[X]$ · $\text{Var}(X)$ · Standard Deviation · Covariance

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Joint & Marginal Distributions

Independence · Marginalization · Conditioning

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MLE & Applications in ANN

Loss functions · Softmax · Why it matters in DNNs



SECTION 01

Probability Foundations

Sample space · Events · Axioms

01

What is Probability?

Definition

A number between 0 and 1 that measures the likelihood of an event occurring. 0 = impossible, 1 = certain.

Classical Formula

$$P(A) = \frac{|\text{Favorable Outcomes}|}{|\text{Total Outcomes}|}$$

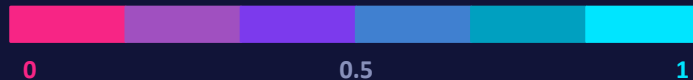
Example: Fair Coin

$$P(\text{Heads}) = 1/2 = 0.5$$

$$P(\text{Tails}) = 1/2 = 0.5$$

$$P(\text{Heads}) + P(\text{Tails}) = 1 \quad \checkmark$$

Probability Spectrum



0.00

Impossible Event

0.10

Very Unlikely (10%)

0.50

Equally Likely / Unlikely

0.90

Very Likely (90%)

1.00

Certain Event

Sample Space & Events

Ω Sample Space

The set of ALL possible outcomes of an experiment.

Dice: $\Omega = \{1, 2, 3, 4, 5, 6\}$

Coin: $\Omega = \{H, T\}$

A, B Events

A subset of the sample space. An event occurs if the outcome is in the subset.

Event A = 'even number' = $\{2, 4, 6\}$

$\emptyset$ Impossible Event

An event that cannot occur. It has zero probability.

$P(\emptyset) = 0$ always

Kolmogorov's 3 Axioms of Probability

I

Non-negativity

$P(A) \geq 0$ for every event A

II

Normalization

$P(\Omega) = 1$ (something must happen)

III

Additivity

If $A \cap B = \emptyset \rightarrow P(A \cup B) = P(A) + P(B)$



SECTION 02

Probability Rules

Complement · Addition · Multiplication

02

Core Probability Rules

Complement Rule

$$P(A^c) = 1 - P(A)$$

The probability that A does NOT occur equals 1 minus the probability that A occurs.

Addition Rule (General)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Mutually Exclusive ($A \cap B = \emptyset$):

$$P(A \cup B) = P(A) + P(B)$$

Multiplication Rule

$$P(A \cap B) = P(A) \cdot P(B | A)$$

For INDEPENDENT events — $P(B | A) = P(B)$:

$$P(A \cap B) = P(A) \cdot P(B)$$

Example: Two fair coin flips

$$P(H \cap H) = 0.5 \times 0.5 = 0.25$$

Quick Example: Deck of Cards

$$P(\text{Red}) = 26/52 = 0.5 \quad \cdot \quad P(\text{Queen}) = 4/52 \quad \cdot \quad P(\text{Red Queen}) = 2/52$$

$$P(\text{Red} \cup \text{Queen}) = P(\text{Red}) + P(\text{Queen}) - P(\text{Red} \cap \text{Queen}) = 26/52 + 4/52 - 2/52 = 28/52 \approx 0.538$$

SECTION 03

Conditional Probability & Bayes

$P(A|B)$ · Law of Total Probability · Bayes' Theorem

Conditional Probability $P(A|B)$

$$P(A | B) = P(A \cap B) / P(B) \quad , \quad \text{where } P(B) > 0$$

The probability of event A, given that event B has already occurred.

Medical Test Example

A patient tests positive (+). What is the probability they actually have the disease?

$$P(\text{Disease} | \text{Positive}) = P(\text{Disease} \cap \text{Positive}) / P(\text{Positive})$$

This is more useful than $P(\text{Positive} | \text{Disease})$!

Spam Filter Example

An email contains the word 'FREE'. What is the probability it is spam?

$$P(\text{Spam} | \text{'FREE'}) = P(\text{Spam} \cap \text{'FREE'}) / P(\text{'FREE'})$$

Conditional probability powers Naive Bayes classifiers.

Bayes' Theorem

$$P(A | B) = [P(B | A) \cdot P(A)] / P(B)$$

Or equivalently: $P(A | B) = P(B | A) \cdot P(A) / [P(B|A) \cdot P(A) + P(B|A^c) \cdot P(A^c)]$

$P(A|B)$

**Posterior
Probability**

Updated belief about A after seeing
B

$P(B|A)$

Likelihood

How likely is B if A is true?

$P(A)$

**Prior
Probability**

Initial belief about A before evidence

$P(B)$

**Marginal
Evidence**

Total probability of observing B

SECTION 04

Random Variables & Distributions

PMF · PDF · CDF · Bernoulli · Binomial · Gaussian

04

Random Variables

X Random Variable

A function that maps outcomes of a random experiment to numerical values.

$$X: \Omega \rightarrow \mathbb{R}$$

Example: X = number of heads in 3 coin flips

Discrete RV

Takes countable values: 0, 1, 2, 3, ...

Described by a Probability Mass Function (PMF):

$$P(X = x)$$

Examples: Number of emails, defects in a batch

Continuous RV

Takes uncountable values in a range $[a, b]$

Described by a Probability Density Function (PDF):

$$f(x)$$

Examples: Height, weight, neural activations

CDF: $F(x) = P(X \leq x)$ — cumulative probability up to x , defined for both discrete and continuous RVs.

Key Distributions in ANN

Bernoulli Distribution

$$P(X=1) = p, \quad P(X=0) = 1-p$$

Models a single binary trial: success or failure.

Used in: Binary classification output (sigmoid).

$$E[X] = p$$

$$\text{Var}(X) = p(1-p)$$

Binomial Distribution

$$P(X=k) = C(n,k) \cdot p^k \cdot (1-p)^{n-k}$$

Models k successes in n independent Bernoulli trials.

Used in: Ensemble models, dropout probability.

$$E[X] = np$$

$$\text{Var}(X) = np(1-p)$$

Gaussian (Normal) Distribution

$$f(x) = (1/\sigma\sqrt{2\pi}) \cdot \exp(-(x-\mu)^2/2\sigma^2)$$

The bell curve. Most common in nature and ANN.

Used in: Weight init, noise modeling, latent spaces.

$$E[X] = \mu$$

$$\text{Var}(X) = \sigma^2$$

SECTION 05

Expectation & Variance

$E[X]$ · $\text{Var}(X)$ · Standard Deviation · Covariance

05

Expected Value & Variance

$E[X]$ — Expected Value

Discrete: $E[X] = \sum x \cdot P(X=x)$

Continuous: $E[X] = \int x \cdot f(x) dx$

The 'center of mass' of the distribution.
Weighted average of all possible values.

$\text{Var}(X)$ — Variance

$$\text{Var}(X) = E[(X-\mu)^2] = E[X^2] - (E[X])^2$$

Measures how spread out values are around the mean. Always ≥ 0 .

σ — Standard Deviation

$$\sigma = \sqrt{\text{Var}(X)}$$

Square root of variance. Same units as X . More interpretable than variance.

Useful Properties & Covariance

- $E[aX + b] = a \cdot E[X] + b$
- $\text{Var}(aX + b) = a^2 \cdot \text{Var}(X)$
- $\text{Cov}(X, Y) = E[(X-\mu_x)(Y-\mu_y)] = E[XY] - E[X] \cdot E[Y]$
- If X, Y independent $\rightarrow \text{Cov}(X, Y) = 0$ and $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$

SECTION 06

Joint & Marginal Distributions

Independence · Marginalization · Conditioning

Joint, Marginal & Conditional

Joint Distribution $P(X,Y)$

Describes the probability of two random variables simultaneously.

Discrete: $P(X=x, Y=y)$

Continuous: $f(x, y)$

Required for multivariate analysis.

Marginal Distribution $P(X)$

Obtained by summing/integrating over the other variable.

$$P(X=x) = \sum_y P(X=x, Y=y)$$

$$f(x) = \int f(x, y) dy$$

Independence

X and Y are independent if:

$$P(X,Y) = P(X) \cdot P(Y)$$

$$f(x,y) = f(x) \cdot f(y)$$

Key assumption in many ANN models.

Conditional Distribution

$$P(X \mid Y=y) = P(X,Y=y) / P(Y=y)$$

Slice of the joint distribution given one variable is fixed.

Foundation for Bayesian inference.



SECTION 07

MLE & Applications in ANN

Loss functions · Softmax · Cross-Entropy · Why probability matters in DNNs

07

Maximum Likelihood Estimation (MLE)

$$\theta_{\text{mle}} = \operatorname{argmax}_{\theta} L(\theta; X) = \operatorname{argmax}_{\theta} \prod_i P(x_i | \theta)$$

Find the model parameters θ that make the observed data most probable. In practice we maximize the log-likelihood.

Log-Likelihood (Practical Form)

$$\ell(\theta) = \log L(\theta; X) = \sum_i \log P(x_i | \theta)$$

Products $\rightarrow$ sums. Numerically stable.

Maximizing ℓ = minimizing negative ℓ (the loss!)

Cross-Entropy Loss (MLE in Disguise)

$$H(y, \hat{y}) = - \sum y_i \cdot \log(\hat{y}_i)$$

Minimizing cross-entropy = maximizing log-likelihood of labels given model output.

Probability Concepts $\rightarrow$ ANN Training

Softmax

Converts logits to a probability distribution over classes (sums to 1)

Dropout

Randomly zeros neurons with probability p — a Bernoulli random variable

Weight Init

Gaussian-initialized weights: $W \sim N(0, \sigma^2)$ prevents vanishing/exploding gradients

Variational AE

Encoder learns μ and σ of a Gaussian latent distribution

Summary

- Probability quantifies uncertainty — essential for building reliable AI systems.
- Conditional probability and Bayes' theorem update beliefs given new evidence.
- Distributions (Gaussian, Bernoulli, Binomial) model real-world data in neural nets.
- Expected value and variance characterize the behavior of random variables.
- MLE directly connects probability theory to training neural networks via loss functions.
- Understanding probability is the foundation for understanding why ANN training works.



Thank You

for your attention

Main Concepts of Probability · ANN Course Project